Practical - 2

Aim: Given a sorted array and a target value return the index if the target is found. Ifnot return the index where it would be if it were inserted in order.

Source Code:

```
#include <stdio.h>
int searchInsert(int* nums, int numsSize, int target) {
    int left = 0, right = numsSize - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) {
            return mid;
        } else if (nums[mid] < target) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }
    return left;
}
int main() {
    int n, target;
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);
    int nums[n];
    printf("Enter %d sorted elements: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }
    printf("Enter the target value: ");
    scanf("%d", &target);
    int index = searchInsert(nums, n, target);
    printf("The index is: %d\n", index);
    return 0;
}
```

Output:

```
Enter the number of elements in the array: 4
Enter 4 sorted elements: 45
62
78
35
Enter the target value: 65
The index is: 2


=== Code Execution Successful ===
```